THADOMAL SHAHANI TSEC ENGINEERING COLLEGE

Assignment 11

Aim. Implement a c program using recursion to calculate the factorial of a given number and display all the factorials from 1 to n

Theory

Recursion is a programming technique where a function calls itself to solve smaller instances of the same problem

Factorial function

if n=0 or n=1 then it returns 1

otherwise it returns n x factorial(n-1)

~~Before that it prints the numbers~~

~~There~~ Print factorials

It has a current and a number to end. It calls factorial starting from 1 to n ~~and recursively~~ and prints it.

It recursively ~~prints~~ calls itself ~~to~~, increasing the start by 1.

Conclusion

We have daisy chained recursion and calling a recursive function from another recursive function and how to efficiently manage it all to give a desired output.

```c
#include <stdio.h>

int factorial(int n) {
   if (n == 0 || n == 1)
      return 1;
   return n * factorial(n - 1);
}

void printFactorials(int current, int n) {
   if (current > n)
      return;
   printf("%d! = %d\n", current, factorial(current));

   printFactorials(current + 1, n);
}

void main() {
   int n;
   printf("Enter a number: ");
   scanf("%d", &n);
   printFactorials(1, n);
}
```

```
Enter a number: 5
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
```